Coffee Sales Analysis Using Microsoft Excel

From Raw Data to Interactive Dashboard

10,388	497	28	14
Sales records	Customers	Products	Cities

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Project Overview

Business problem

A coffee retailer sells across 14 cities and 28 products, but data sits in four separate tables. Which products, cities and periods drive revenue, and how satisfied are customers?

Objectives

- Prepare one clean analytical dataset
- Answer business questions with formulas
- Report through an interactive dashboard

1

Power Query

Import and consolidate Sales, Customers, Products and City

2

Formulas & dynamic arrays

XLOOKUP, FILTER, UNIQUE, SORTBY, LET, SUMIFS

3

Formula-driven dashboard

No PivotTables and no VBA by design

Dataset Overview

10,388

Sales table

sale_id, sale_date, product_id, customer_id, total, rating

497

Customers table

customer_id, customer_name, city_id, city_name, sales

28

Products table

product_id, product_name, price

14

City table

city_id, city_name, population, estimated_rent, city_rank

10,927 source records · Sales dates 1 Jan 2023 to 1 Oct 2024 · Total sales ₹60,70,190

Data Preparation

1. **Import**
   Load the four source CSVs with Power Query
2. **Consolidate**
   Combine the sheets into one analytical dataset
3. **Structure**
   Keep each source as a named Excel table
4. **Enrich**
   Add city_name to customers with XLOOKUP

XLOOKUP enrichment

```
=XLOOKUP(
   [@[City_Id]],
   city[city_id],
   city[city_name])
```

City_Id stays as the foreign key; city_name is added as a readable column for reporting.

Data Quality Checks

✓ No blanks

No null values in the four source tables

✓ No duplicates

No duplicate rows or primary-key IDs

✓ Keys match

Customer city_id, sales product_id and customer_id all resolve

✓ Dates parse

Sales dates run 1 Jan 2023 to 1 Oct 2024

✓ Totals consistent

All totals positive; each equals the product price

✓ Ratings valid

Ratings range 2 to 5, inside the 1 to 5 scale

To re-run after the final transformations and before publishing.

Excel Functions and Techniques

Function	Used for	Example
XLOOKUP	Map city IDs to city names	`=XLOOKUP([@[City_Id]],city[city_id],city[city_name])`
FILTER + UNIQUE	Records over a threshold; distinct cities	`=FILTER(customers[sales],customers[sales]>1000)`
LARGE / SMALL + SEQUENCE	Top 3 prices above 1,000	`=LARGE(FILTER(products[price],products[price]>1000),SEQUENCE(3))`
SUMIF / SUMIFS	City totals, with a sales band	`=SUMIFS(customers[sales],customers[city_name],c,customers[sales],">=500")`
LET + CHOOSE	Two-column dynamic summaries	`=LET(c,UNIQUE(customers[city_name]),s,SUMIF(...),CHOOSE({1,2},c,s))`
SORTBY	Sorted, ranked outputs	`Rank cities and products`

Key Performance Indicators

₹60,70,190 Total sales	**10,388** Orders	**₹584** Average order value	**3.99** Average rating
497 Customers	**28** Products	**14** Cities	**Q4 2023** Peak quarter, ₹19.6 lakh

Calculated from the sales table: total ÷ order count gives the average order value.

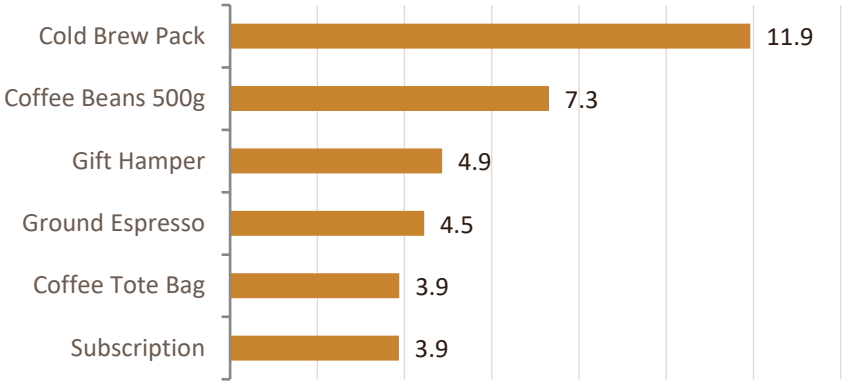
Interactive Excel Dashboard



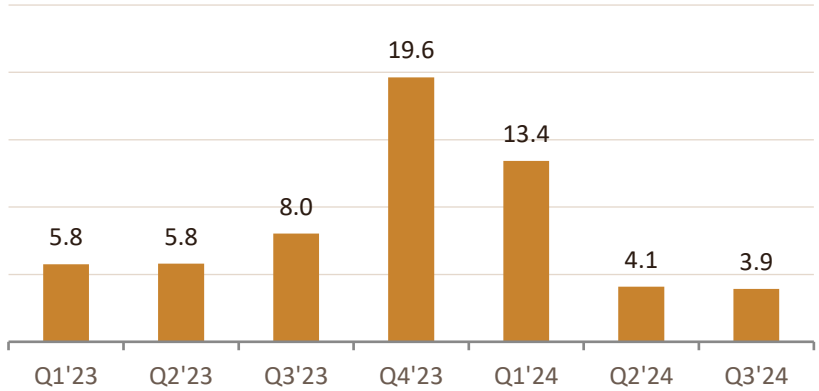
Pick a filter → formulas recalculate → KPIs and charts update (dropdowns: next build step)

Key Findings and Business Insights

Top 6 products by revenue (₹ lakh)



Quarterly revenue (₹ lakh)



Concentrated

Top 3 products bring about 40% of ₹60.7 lakh

Q4 2023 peak

₹19.6 lakh, 32% of all revenue, then a steep fall

Satisfied base

Average rating 3.99; 74% of ratings are 4 or 5

Conclusion and Project Access

Skills demonstrated

- Data preparation with Power Query
- Enrichment with XLOOKUP
- Dynamic-array analysis and validation
- Dashboard reporting and business insight

About me and links

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